Autrin Hakimi

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PROFESSIONAL PROFILE

Software engineer building **full-stack applications**, **ML-driven systems**, and **robotics automation**. Currently at ASM, where I independently own and deliver multiple full-stack systems for semiconductor manufacturing. Published researcher (**ACM SIGSPATIAL 2025**) with experience in distributed deep learning across **64 NVIDIA A100 GPUs**. Strong across Python, C/C++, JavaScript, and SQL, with end-to-end ownership from design through production.

TECHNICAL SKILLS

Languages Python, C/C++, JavaScript, TypeScript, SQL, Bash, HTML, CSS
Web & Backend React, Python Flask, Tkinter, REST APIs, WebSockets, JavaScript/HTML/CSS
Cloud & DevOps Docker, Git, CI/CD pipelines, PyInstaller/NSIS packaging, Linux/Unix, Microsoft Azure, Google Cloud, Snowflake
ML & Data PyTorch, scikit-learn, Pandas, NumPy, distributed training (64-GPU), DGL, NetworkX
Databases MySQL, SQL, SQLite, NoSQL, Snowflake
Automation & Control Robot workflow orchestration, vision systems, HMI/GUI, ROS, OpenCV, Yamaha robot controllers
Methodologies Agile (Scrum/Kanban), test-driven development, cross-functional collaboration

EDUCATION

B.S., Computer Science **Aug 2021 - May 2025**
Iowa State University | GPA: 3.70/4.0 | Magna Cum Laude | Dean's List *Ames, IA*

PROFESSIONAL EXPERIENCE

Software Engineer **Jan 2026 - Present**
ASM America *Phoenix, AZ*

- Independently own and deliver multiple **full-stack production systems** for semiconductor manufacturing: robotics platforms, an ML-driven parts matching tool, robotics frontends, and operator tooling; work that typically requires multiple engineers.
- Inherited and overhauled a legacy codebase, redesigning architecture across both backend and frontend; rebuilt the UI of a **collaborative robot application** shipping with a seven-figure robotics product, contributing to a **quarterly team award**.
- Resolved numerous backend issues spanning **synchronization, interrupt handling, robot stop conditions, and measurement algorithms**; improved measurement accuracy and built automated validation suites (**140+ tests**) verifying geometry and control logic off-hardware.
- Built a full-stack **ML-driven parts matching system** (Flask, JavaScript, SQL) that saves engineers hours of manual lookup; engineered **threading and synchronization** across robotics frontends to coordinate vision and control workflows.
- Package and deploy applications to production systems at **international manufacturing sites** (PyInstaller, NSIS); provide remote troubleshooting and field support for globally deployed tooling.
- Lead **requirements definition and stakeholder alignment** for digitizing a customer-facing qualification workflow; introduced **Git** and version control practices to the team; collaborate cross-functionally with mechanical, electrical, and process engineering.

Undergraduate AI Researcher **Aug 2024 - Nov 2025**
Iowa State University, SwAPP Lab *Remote*

- Co-authored “HydroGAT” [paper, code] accepted at **ACM SIGSPATIAL 2025**: a heterogeneous graph attention transformer for flood prediction reaching **NSE 0.97**; built distributed training across **64 NVIDIA A100 GPUs** on NERSC Perlmutter with **15× speedup**.
- Engineered end-to-end **data pipelines** (PyTorch, DGL, Pandas, NumPy) for large-scale datasets; automated workflows via Bash scripting, cutting setup time **40%**.

Software Development Intern **May 2024 - Aug 2024**
BuilderTREND *Omaha, NE*

- Designed and shipped a **React**-based job proposal template; developed full-stack features in JavaScript, TypeScript, and C# (.NET).
- Migrated legacy ASP pages to React, improving scalability; collaborated in **Kanban/Scrum** sprints with cross-functional teams.

Data Science Intern **May 2023 - May 2024**
Alliant Energy *Remote*

- Developed **predictive ML models** (Random Forest, SVM) in Python and SQL analyzing electricity usage for **995K customers** in Snowflake.
- Predicted EV ownership for customer segmentation; recognized by Director of Data Analytics for impactful business insights.

SELECTED PROJECTS

Autonomous Robotic Chess System [code]

- Built fully autonomous system: real-time piece recognition (**OpenCV**), move calculation, and **UR10e robot arm control** (ROS, MoveIt, Python).

xv6-RISC-V OS Schedulers [code]

- Implemented **Round-Robin and Stride schedulers** in xv6 on RISC-V; C, context switching, spin locks, and process synchronization.